Second, a team of researchers at Texas Tech University used a critical path to determine when to take distributions from an investment portfolio and when to take distributions from a reverse mortgage line of credit. I discussed that method in my book *Reverse Mortgages: How to Use Reverse Mortgages to Secure Your Retirement*. Third, David Zolt used the critical path to decide whether or not a retiree should take a full inflation adjustment for spending during each year of retirement. This variable spending strategy was discussed in chapter 6.

The “critical path” helps determine if the portfolio is on target to meeting the retirement goal. First, a retiree determines an end goal for the portfolio in terms of how long the portfolio should last, as well as how much spending it should support. Examples of portfolio end goals could be to deplete the portfolio by age 105, to maintain $100,000 at age 90, and so on. Next, these numbers are combined with the current portfolio value to determine a portfolio return assumption that will allow the end goal for the portfolio to be met. Third, this information is combined to determine a glide path for the value of remaining wealth throughout retirement, showing the wealth needed to remain precisely on track to meet the spending and portfolio end value goals. The critical path compares where the portfolio is and where it should be in dollar terms to be on track.

Volatility is not an appropriate measure of risk for personal financial planning. When one is in the safety zone above the critical path, volatility may not be risky, but a low-volatility portfolio may be quite dangerous when one has fallen below the critical path and the portfolio is plummeting toward zero. Portfolio volatility is not the same as risk; while they are related, the real risk in personal finance is witnessing the wealth level fall below the critical path. In a sense, high volatility when above the critical path is less risky than low volatility when below the critical path.

Exhibit 7.18 provides an example of a critical path for retirement wealth. This individual has $1 million at retirement. She seeks to support $40,000 of spending with a 2 percent COLA for a forty-five-year retirement period. Wealth is allowed to be depleted at the end of forty-five years. She determines that a 5 percent return is needed to support this retirement spending goal with her portfolio. The exhibit tracks the path of her remaining wealth with a 5 percent return to let her know where she should